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Nonverbal Behavior Recognition in Multimodal Interactions from RGB Video **3**
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Effective human collaboration relies heavily on understanding nonverbal behaviors, which provide critical evidence of implied intent and social dynamics that speech alone cannot capture. This chapter introduces a comprehensive RGB audio-visual perception pipeline designed to extract, integrate, and structure both verbal and non-verbal events from multi-party interactions. Our system integrates person and object detection models, multi-target tracking, and detailed analysis of key social signals. These signals include gaze (to detect individual, mutual, and joint attention), gestures (such as pointing, nodding, and head shaking), posture (including leaning in, out, or to the side), and speech and speaker identification. The pipeline processes these rich multimodal cues into coherent, structured event streams, which provide a robust foundation for modeling situational awareness and group dynamics. This chapter also presents a detailed evaluation of the pipeline's performance, focusing on nonverbal recognition, demonstrating its efficacy and reliability in capturing these complex behaviors. We conclude by discussing how these event streams support critical downstream applications, including semi-automatic annotation, engagement tracking, common ground analysis, and friction detection, laying the groundwork for more perceptive AI-driven collaborative systems.

1 Introduction

To understand participants' latent intentions, desires, and ideas in multi-party interaction and to maintain situational awareness, dialogue acts (e.g., chat messages and speech acts) are often insufficient on their own (D. Palmer et al., 2025). Nonverbal behaviors (e.g., facial expression, gaze, gesture, posture, and object movements) provide complementary understanding of the verbal channel (Dey and Puntambekar, 2023). Motivated by the fact that many collaborative problem-solving corpora consist of pre-recorded 2D RGB video with synchronized audio (Echeverria et al., 2017; Khebour et al., 2024; Sarkar, Posen, and Etemad, 2022; Sun et al., 2025), we introduce a 2D audio-visual detection system that extracts and structures both verbal and nonverbal events. The resulting event streams enable reconstruction of the evolving situation, and therefore support situational awareness, enhancing capabilities in perception, comprehension, and prediction and enabling downstream tasks such as semi-automatic annotation and analysis, engagement tracking, and interaction with an AI agent (D. Palmer et al., 2025).

2 Related Work

Research on collaborative problem-solving (CPS) investigates problem-solving skills from multiple perspectives in socially rich contexts and evaluates the dynamics of collaborative processes (Nelson, 2013). Within this setting, situation awareness (SA), a person's perception and comprehension of task-relevant information in a scenario, is critical for effective communication and task success (Endsley, 2021).

To support CPS and SA, techniques in multimodal social signal processing have enabled automatic analysis of verbal and nonverbal behaviors (Wagner et al., 2013). Multimodal social signal extraction and processing integrate multiple communicative cues, offering insights into engagement and group dynamics. Progress has been made in extracting and analyzing social signals across multiple modalities. Speech conveys explicit meaning, as well as prosodic cues of intention

(Chiu, 2008; Shriberg and Stolcke, 2004). Gaze reflects attention targets, mutual attention, and joint attention (Fathi, Li, and Rehg, 2012; Huang et al., 2015). Posture provides cues about group engagement and collaboration (Miller, 2005; Radu, Tu, and Schneider, 2020). Gesture carries meaning in itself or enhances the meaning of verbal acts (Brutti et al., 2022; Komorowska, 2017). Facial expression overtly communicates belief states, agreement, or doubt (Miller, 2005). Taken together, these modalities provide a comprehensive basis for understanding and modeling collaborative dynamics in multi-party interaction corpora.

3 Pipeline

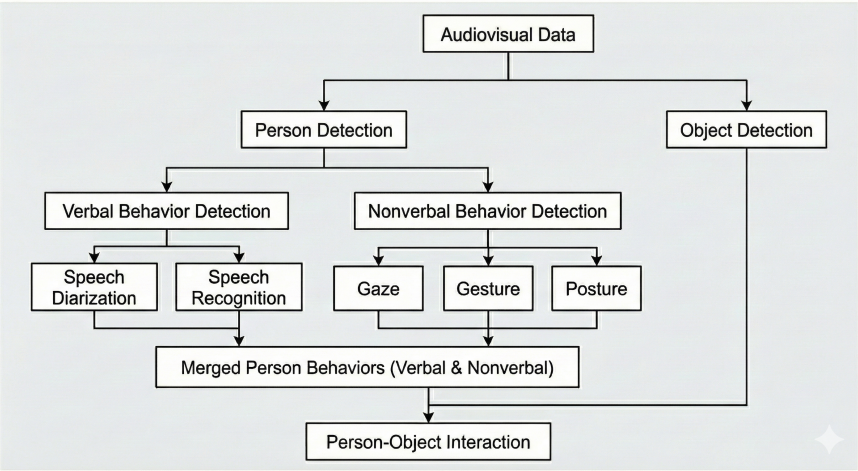


Figure 0.1: Flowchart of detection pipeline

This pipeline processes RGB video or pre-extracted frames with audio files to extract both verbal and nonverbal behaviors by integrating person and object detection with multi-target tracking of the detected entities, as shown in Figure 0.1. For each detected person, it estimates posture, gestures, gaze, facial expressions, and speech acts from the audio–visual stream. It also tracks object state changes to infer

implied actions (e.g., turning the laptop screen). In human–human interaction, the system detects resource sharing (e.g., using the same tool) and co-attention (mutual gaze); in human–object interaction, it recognizes actions such as typing and identifies joint-attention events, where two or more people attend to the same object.

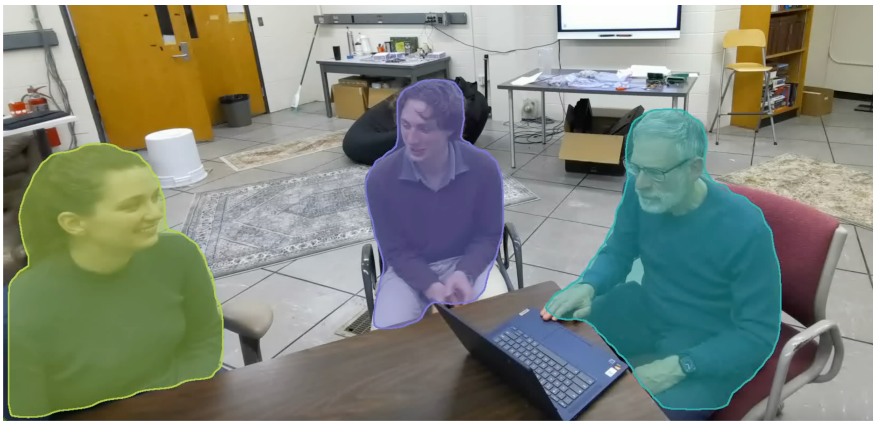
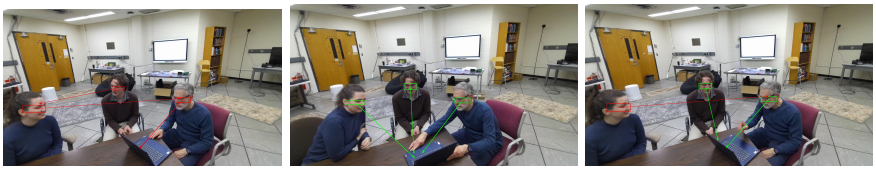


Figure 0.2: The pipeline detects three individuals in the frame, and each person is segmented for visualization.

3.1 Person Detection

We use Detectron2 (Wu et al., 2019) to detect persons in each frame. The resulting detections are associated with active TransT (X. Chen et al., 2021) tackers using IoU matching: matched tracks are updated through the tracker's `track` routine, while unmatched detections initialize new trackers. For each detected person, a top-down MMPose estimator (MMPose Contributors, 2020) is applied to infer body keypoints for nonverbal behavior analysis. Figure 0.2 shows the detected individuals along with their segmentation within a frame.



(a) Red gaze line: individual attention.

(b) Green gaze line: joint attention.

(c) Individual attention and joint attention in the same scenario

Figure 0.3: Illustration of three distinct gaze visualization scenarios.

Gaze

Most RGB video datasets, including the one used in this chapter, do not provide depth information. In addition, participants are not equipped with eye-tracking devices, and thus no ground-truth eye gaze annotations are available for supervised training. Accordingly, we approximate gaze direction using head orientation as a proxy. This design choice is motivated not only by data constraints but also by practical and ethical considerations. In many human-related video analysis, faces must be blurred to protect participants' privacy, and privacy regulations often preclude the use of identifiable facial features. Under such conditions, head pose provides a more practical and ethically compliant basis for gaze inference. Moreover, equipping participants with eye-tracking hardware in multiparty collaborative settings is often prohibitively costly and logistically infeasible, particularly when scaling to larger groups.

Inputs. The gaze module takes two inputs: (i) a JSON file mapping each frame identifier to a list of per-frame head bounding boxes, and (ii) the corresponding RGB frames extracted from the source video. Head bounding boxes are generated by the detection and tracking pipeline described above, but restricted to the head region. For each tracked individual, we extract five landmarks (nose, left/right eye, left/right ear). To support downstream tasks, each bounding box is serialized in normalized image coordinates $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$,

and the results are stored in a JSON file. Video frames are obtained by sequentially decoding the stream with `cv2.VideoCapture` and saving every n-th frame as zero-padded images, ensuring that lexicographic sorting preserves temporal order. All images are saved in a designated image folder.

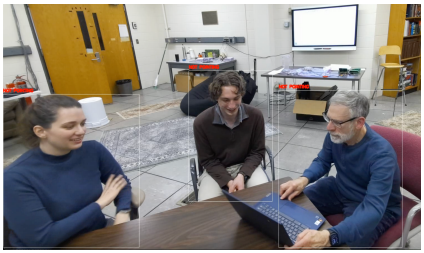
Model and inference. Given the image folder, the Gaze-LLE pipeline (Ryan et al., 2025) iterates over the sorted frames. For each frame, it loads the corresponding normalized head boxes from the JSON file and forwards the frame through a pretrained gaze-target predictor. The model outputs a per-head heatmap over potential gaze targets; the peak location is then identified and mapped back to the original resolution to yield the gaze pixel target (g^x, g^y) for each head.

Visualization. For qualitative analysis, we overlay three elements on each frame: (i) the head bounding box for each individual, (ii) a point marking the predicted gaze target, and (iii) a line segment connecting the head box to its target. By default, these overlays are in red to indicate individual gaze. When two individuals have co-attention (mutual gaze), where each target falls within the other's head box, their overlays are changed into blue. When two or more individuals share the same gaze target (human or object), the corresponding overlays are updated into green. Figure 0.3 illustrates different gaze scenarios in a real interaction setting.

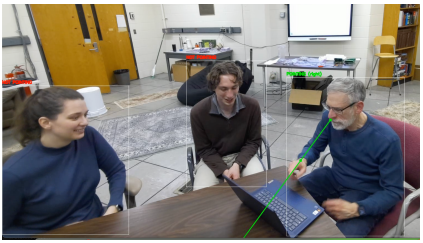
Pointing Gesture

For the present discussion, we focus only on pointing gestures in this chapter. While other kinds of gestures, such as iconic and emblematic are semantically meaningful, they are not addressed as they are very rare in our dataset.

The pointing detection module is built on YOLOv8 pose estimation (Jocher, Qiu, and Chaurasia, 2023) to extract upper-body keypoints for each detected person in a video stream. Arm configurations are



(a) No pointing detected: all three participants are labeled in red.



(b) Pointing detected for the participant on the right, indicated by a green label.

Figure 0.4: Illustration of the pipeline's pointing detection: (a) absence of pointing and (b) detection of a pointing event.

then analyzed to determine whether an arm is extended in a pointing gesture. The decision is based on several features: elbow angle, arm extension relative to shoulder width, horizontal displacement, elevation, and torso-relative orientation. A hip-based proxy is also used to suppress false positives when wrists fall below desk height, which typically indicates resting or typing.

For qualitative analysis, each detected person is marked with a bounding box and label, with "NOT POINTING" shown in red and "POINTING" shown in green. When pointing is detected, an overlay is drawn from the shoulder to the wrist, and a projected line is extended to indicate the pointing direction. These two scenarios are illustrated in Figure 0.4.

Posture

The pipeline focuses on two types of posture change: leaning in or out, and leaning to the right or left. For each participant, posture is estimated from body keypoints around the neck, hips, and shoulders. Leaning in or out is determined by changes in the apparent torso scale relative to a neutral baseline established at the start of the video, with increases interpreted as leaning in and decreases as leaning out.

Lateral lean is inferred from the angle of the shoulder line, with a consistent rightward tilt indicating a lean to the right and a leftward tilt indicating a lean to the left. Movements that do not exceed these thresholds are considered neutral.

3.2 Object Detection and Movement Tracking

Inputs. The object detection module takes an RGB video stream together with a user-specified set of task-relevant object classes (e.g., *laptop*, *keyboard*, *mouse*) as input.

Model and inference. We use MMDetection (K. Chen et al., 2019) to identify task-relevant objects in each frame. Detected objects are tracked and matched across frames using greedy IoU matching, which maintains continuous tracks and allows the system to capture their positions and movement over time. An object is considered *moving* when its location shifts noticeably or when its overlap with the previous frame's bounding box decreases substantially.

Visualization. Objects identified as moving are highlighted in green, as illustrated in Figure 0.5.

3.3 Speech and Speaker ID Detection

For all recorded audio inputs, when the primary objective is to identify the speaker in order to assess whether a given speech act co-occurs with nonverbal behaviors, such as nodding by addressees, pointing by the speaker, or shifts in attention from interlocutors, we use TalkNet (Tao et al., 2021), an audio-visual active-speaker detection model that integrates audio and visual modalities to determine whether a given face in frame is speaking. By contrast, when both speaker identification and aligned transcripts are required, we use a two-stage pipeline: PyAnnote (Bredin et al., 2020) for speaker diarization, and then we apply Whisper (Radford et al., 2023) on diarized results

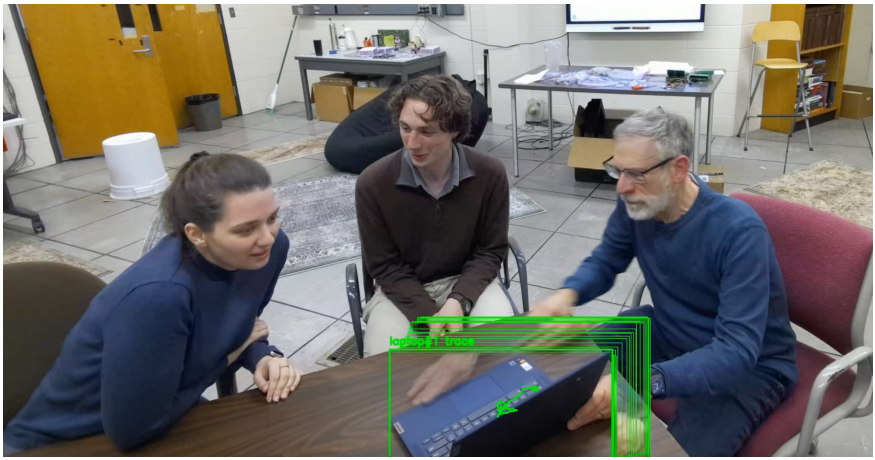


Figure 0.5: The laptop is rotated from facing the participant on the right toward the rest of the group. The green bounding box marks the detected laptop and its turning trajectory, while the curly line inside illustrates the trace of its motion.

and the audio stream to generate transcripts annotated with speaker identity.

4 Evaluation

4.1 Study Context

Participants completed a jigsaw activity adapted from a broader curriculum. In the jigsaw approach (Aronson et al., 1978), students first develop expertise in a specific topic and then collaborate to integrate their knowledge to solve a shared problem. In our study, participants learned about one of three sensors (environmental, moisture, or sound), and then worked collaboratively (in groups of 2-3 participants) to share knowledge, answer questions, and brainstorm solutions to real-world problems that required integration of all three sensors.

The study was conducted in a laboratory setting. Each session was video recorded using a mounted camera facing the participants as well as computer recording; the computer screen was also recorded to capture participants' activity.

Video Selection

We selected video recordings of two groups, each consisting of three students working on this activity. These groups were chosen because they exhibited high levels of interaction between them. From the recordings, we selected 12 video clips (ranging from 10-25 seconds, with an average duration of 14 seconds, at 60 FPS), depicting multiple students simultaneously exhibiting multiple nonverbal behaviors.

Video Angles Used for Analysis

To annotate the videos, human annotators watched the videos of the students participating in the collaborative activities. Each recording displayed three synchronized video streams: two were camera views of the participants and the third was a screen recording of their computer activity. The primary camera view provided a slightly elevated frontal perspective capturing all the participants, allowing annotators to observe mostly their gaze, gestures and interactions with the tools, and postures. There are other instances, where the annotator was unable to get an exact assessment of what is transpiring. For instance, glasses, cups, or participants' hair, will obstruct the gaze direction or certain gestures and often these gestures will be even more difficult to determine from a specific angle. In such cases, the annotators will have to seek the second angle to get certain information from that perspective. However, this secondary view often did not capture the side participants' arms or faces, especially if they leaned back slightly. Thus, the elevated frontal view was used as the primary source for annotation, while the secondary view used to resolve ambiguity. Consequently, this primary view was also used as the video source for the pipeline.

The screen recording was used to identify usage of shared resources. However these behaviors were not included in the pipeline as it is difficult to reliably distinguish between "hand on tool" and active tool use from the available camera angle (the primary camera view). For example, a student may have their hand on the mouse while actively clicking on something on the screen or simply moving the cursor around. As a result, videos and annotations related to tool use were excluded from the pipeline and are beyond the scope of this study.

These additional video angles were primarily used to resolve edge cases during human annotation. We specifically chose certain edge cases to determine how the model performs in these ambiguous situations, and to understand how they might need to be accounted for going forward.

Video Processing

The pipeline takes synchronized video–audio data as input. It first decodes the video into individual frames, then detects relevant persons and objects. Based on these detections, the system predicts a range of nonverbal and verbal behaviors. The outputs include structured predictions in JSON format, corresponding visualizations, and high-level summary reports for each modality.

4.2 Nonverbal Interactions in Collaborative Environments (NICE) framework

We leveraged the Nonverbal Interactions in Collaborative Environments (NICE) framework (Dey and Puntambekar, 2023) to evaluate the pipeline performance. The NICE framework characterizes students' nonverbal behaviors during small-group collaboration across three modalities: gaze (e.g., joint attention, looking at each other, or looking away from the group), gesture and action (e.g., pointing and tool use), and body motion (e.g., head nods and leaning). Annotating videos using the NICE framework enables analysis of the extent and

nature of each group members' nonverbal participation, providing insight into the collaborative process.

Certain nonverbal behaviors are indicative of students' attention and focus during collaboration and can therefore inform our understanding of their level of participation in collaborative activity (Barron, 2003; Tomasello, 2014). The direction of eye gaze is an important component of the NICE framework. It captures whether students are visually focused on the same object at the same time, referred to as *Joint Visual Attention* (Schneider and Pea, 2015), whether they look at one another during discussion, maintain individual focus, or look away from the activity area altogether. Looking at others while speaking or listening, may be indicative of respect and also help groups establish rapport (Rogat and Linnenbrink-Garcia, 2011). Individuals may also focus on their own work to complete the task but not interact as much. Such ongoing parallel interactions (Forman and Cazden, 2013) are indicative that the group is not collaborating well. *Looking away* from the group activity area (e.g., looking around the classroom, looking out the window) for longer periods of time may be indicative of disengagement from the activity.

To account for the frequent use of tools and shared resources in computer-supported collaborative learning environments, the NICE framework distinguishes between collaborative and individual tool use. In addition, deictic gestures, such as pointing at tools or at other group members to direct attention or focus (Evans et al., 2011), support the establishment of shared understanding (Roschelle and Teasley, 1995) and have been associated with collaboration quality (Grover et al., 2016) and therefore are included in the NICE framework.

Body-based behaviors further signal attention and engagement. For example, "huddling" or leaning in together, has been identified as indicating mutual attention and shared interest (Barron, 2003). Responding to others through head nods or shakes often signals nonverbal agreement or attentiveness (Ogden, 2000). Thus, the NICE framework also includes *head nods/shakes* and *leaning toward* as indicators of attention, and *leaning away* as an indicator of disengagement.

For this study and evaluating the pipeline, we used the categories adapted from the NICE framework shown in Table 0.1.

Modality	Annotation category	Description	What is this indicative of?
Gaze	Joint Visual Attention	More than one group member focusing on the same object at the same time	What students are paying attention to.
	Look at others	Looking at others while speaking, or looking at others while they are speaking	Looking at others can also help groups establish rapport.
	Look away	Looking outside group activity area	Looking away may be indicative of disengagement.
	Individual attention	Having individual focus, e.g., looking at their own notebook while others are working together on the computer	
Gesture	Point at tool	Pointing at objects used in the activity	Indicating or directing attention or focus. Can also help establish shared
	Point/gesture at others	Pointing at or indicating others	
Posture	Lean towards	Leaning in towards activity area	understanding. Attention or focus.
	Lean away	Leaning out, away from activity area	

Table 0.1: Categories from the NICE framework used in the pipeline evaluation and their descriptions.

We omitted the tool use annotation categories because it was difficult to reliably distinguish actual tool use from hands resting on objects from the videos. For example, students often rested their hands on the keyboard (hands on object) while discussing next steps with group members, without actively typing or otherwise using the tool. In addition, head nods and head shakes were excluded as they were rarely observed.

4.3 Unit of Analysis and Transcript Segmentation

The unit of analysis in the NICE framework is at the utterance level. That is, the framework helps capture what each student is doing while someone in the group is speaking. Since utterances can vary in length, NICE applies binary annotation (1=present, 0=absent) to indicate whether each behavior is observed (or not) within each utterance, for each individual. For brief pauses between utterances, any changes in nonverbal behaviors are included with the preceding utterance.

Annotations were anchored to utterances and were completed for each participant. These then served as the "ground truth" annotations against which the model performance was evaluated.

We used Whisper to initially generate low fidelity diarized transcripts. A team of transcribers then used these transcripts as a guide to produce medium fidelity transcripts with more accurate speaker identification and transcription, organizing utterances by turn of talk.

Transcript Use by NICE vs Model

Transcription and conversation analysis are traditionally conducted one speaker at a time, under the assumption that speakers alternate turns between speaking and listening, waiting for another speaker to complete a turn before speaking (Sacks, Schegloff, and Jefferson, 1974). However, simultaneous speech is relatively common, particularly in group conversations, and therefore need to be accounted for in transcription and annotation. One form of overlapping speech consists of brief vocalizations (e.g., "*yeah*," "*hmm*," "*uh*", etc.) that enable listeners to display their attention to or understanding of a speaker's talk. These vocalizations are commonly referred to as back-channel behaviors (Yngve, 1970). In the medium fidelity transcripts, back-channels produced during an utterance were initially transcribed after that utterance was completed. However, for the purpose of this analysis, utterances were ordered strictly by their chronological occurrence to preserve the temporal sequences of the interactions (described in the following section), and the nonverbal behaviors were annotated only for the duration of the back-channel utterance itself. Other forms of overlap include interruptions (Ferguson, 1977), in which a new speaker takes over the conversation and the original speaker stops speaking, as well as instances in which speakers continue talking with partial temporal overlap. In all such cases, utterances are transcribed as separate units, and nonverbal behaviors are annotated from the onset of each utterance. Since NICE annotations are anchored to utterance boundaries rather than to the turn-taking structure, precise

temporal overlap between utterances does not affect the annotation process using NICE.

In contrast, the pipeline requires a strictly chronological ordering of utterances as it happens, particularly in cases of back-channeling or overlapping speech. During the detection phase, video inputs are decoded into discrete frames, effectively compressing the distinct temporal trajectories of the three individuals into a single, synchronized timeline. Consequently, nonverbal behaviors are predicted simultaneously for all subjects at each frame. However, this strictly chronological processing creates a misalignment with the NICE framework, where overlapping timestamps and the retrospective analysis of behaviors do not adhere to a linear frame-by-frame sequence.

In this study, we do not analyze the generation of the back-channels themselves, but rather consider how these back-channels (whether verbal or nonverbal) may contribute to the collaborative interaction. We focus only on evaluating the multimodal pipeline performance; examining downstream processes is beyond the scope of this study and will be explored in future work.

Transcription Segmentation

Consequently, we need to segment the transcripts by timestamp, ensuring that each utterance fragment appeared in the exact temporal order (Forsyth, Clarke, and Lam, 2008) for the whole group while keeping the utterance time separate for each individual. First we tried the Whisper small model (approximately 244 million parameters) to generate the timestamps for the transcripts, but the prediction results show that the overlapped utterances are hard to be detected by the model. So we manually exchanged the end time of the preceding utterance with the start time of the subsequent utterance to better reflect the intended segmentation. This adjustment was performed 16 times across seven clips, with the average difference between the sections being less than 1 second (*mean=0.78s, SD=0.67s*). For human annotators, such sub-second differences are difficult to distinguish, and the utterance content itself remained unchanged; thus, we considered

these timestamp exchanges acceptable. Nevertheless, because these adjustments slightly altered utterance boundaries, they may introduce slight discrepancies between human annotations and model-based outputs.

When interruptions or back-channel utterances occurred, we split the original utterance at the point of overlap, inserted the interrupting or back-channel utterance, and then resumed the original utterance (where applicable). This segmentation strategy resulted in utterance boundaries that differed slightly from those used for NICE annotations. Accordingly, we reviewed and, where necessary, adjusted the NICE annotations for affected utterances or utterance segments to ensure that nonverbal behaviors were accurately aligned with the corresponding transcript segments.

Aligning Output Formats

To enable comparison between model outputs and NICE annotations, we aligned the frame-level predictions to the NICE utterance-level binary output format, instead of having the machine predicted results count the frequency of each modality. Specifically, for each utterance, as defined by its start and end timestamps, we aggregated predictions across all frames within that interval. If a given modality was detected in at least one frame during the utterance, the utterance-level output for that modality was labeled as 1; otherwise, it was labeled as 0.

4.4 Results and Explanations

Overall, the model performance is consistent across clips from different videos, despite variation in section boundaries as a result of overlapping speech. Although this variation could potentially affect the alignment between model outputs and human annotations, the relatively stable accuracy across the different clips suggests that our approach to utterance segmentation and splitting was robust, and did not substantially affect model evaluation.

The NICE annotation scheme includes categories for nonverbal behaviors that occur infrequently (e.g., pointing) as well as categories that occur frequently (e.g., joint or individual visual attention) as observed in our dataset, because human behaviors are not predictable or static, and especially across different individuals. This results in class imbalance between and within modalities. Although we selected clips with high levels of interaction compared to rest of the videos, certain behaviors naturally occur more often than others, e.g., gaze is continuously present because students must look somewhere, whereas pointing occurs only when needed. Further, there might be moments of intense joint activity between the participants where all modalities are present for a few moments and then revert to their neutral state, thus leading to imbalances within modalities as well across both individual and group level. To account for variations in class distribution, we evaluated performance using both accuracy (Table 0.2) and F1 score (Table 0.3). Overall, the model achieved good accuracy (0.73 ± 0.06) but a notably lower F1 score (0.59 ± 0.09). This discrepancy likely stems from data imbalances or segmentation decisions; consequently, we conducted a detailed review of the video data to investigate the sources of this reduced performance.

Model Performance: Pointing

The model's overall accuracy across all modalities was 0.73 (see Table 0.2). The model performed particularly well on pointing behaviors, both pointing at tools (*mean accuracy=0.89, SD=0.13; mean F1 score=0.74, SD=0.28*) and pointing at others (*mean accuracy=0.93, SD=0.10; mean F1 score=0.77, SD=0.28*). These behaviors were relatively rare in the dataset, indicating that the model was able to detect and differentiate regular hand movements from pointing gestures with high accuracy.

Model Performance: Gaze

In contrast, the model performed least accurately on most gaze-related categories, including joint visual attention (*mean accuracy=0.46, SD=0.22; mean F1=0.37, SD=0.18*), looking at others (*mean accuracy=0.64, SD=0.22; mean F1=0.45, SD=0.18*), and individual attention (*mean accuracy=0.46, SD=0.22; mean F1=0.32, SD=0.15*). The low F1 scores indicate that the model struggled to simultaneously achieve high precision and high recall for these behaviors. To investigate these discrepancies, we reviewed the corresponding video segments and compared NICE annotations with model predictions. We identified the following potential reasons for discrepancy:

(i) Discrepancy in object definitions for joint visual attention: In the NICE framework, joint visual attention is defined as students attending to the same object simultaneously, i.e., students are focused on or paying attention to the same aspect of the task. In this activity, three objects were relevant to the task as thus considered as the "common" object or tool: the computer screen, the mouse, and the keyboard. For example, when one student typed while looking at the keyboard and another observed the screen, the NICE framework annotation classified this as joint attention because both were engaged with the same task-related content. In contrast, the model detected each object separately and therefore classified such instances as individual attention. This mismatch substantially contributed to lower accuracy in gaze categories. We tested this hypothesis by post-editing the model's predictions to treat gaze at the keyboard and gaze at the screen as joint attention on a single object; under this adjustment, joint visual attention accuracy increased to 1.

Additionally, camera angle limitations further affected gaze classification. The computer screen was entirely off-camera and therefore could not be detected by the model. As a result, when students looked at the screen individually, the model sometimes classified this behavior as looking away from the activity area, even though the students were attending to the task. However, when more than one student was looking at the screen at the same time, the model correctly identified

this as joint visual attention, likely due to consistent head orientation patterns across participants.

(ii) Discrepancy in head and body movements: Differences in head and body movement interpretation also contributed to gaze misclassification. In certain instances (e.g., clips 9 and 10) in video 2, one student frequently swiveled in their chair, producing head and body movements. Although the student's head/body orientation suggested they were looking at another group member, closer inspection of the video revealed that their gaze remained on the screen, maintaining joint attention with their group. The NICE annotations captured this distinction accurately, whereas the model misclassified the student's attention, thus reducing accuracy. Although this represents an edge case, it highlights the importance of human evaluation in validating automated behavioral detection models. It also highlights the need to account for uncommon or nuanced behavioral patterns that pretrained models may not be equipped to accurately classify.

Model Performance: Posture

The model performed well in classifying *lean away* (*mean accuracy=0.94, SD=0.09; mean F1=0.77, SD=0.28*), likely because it was relatively rare. Further the few times it did occur in the clips, it involved quite exaggerated changes in posture that may have been easier to detect. In contrast, the model performed less well in instances of (*lean towards* (*mean accuracy=0.62, SD=0.22; mean F1=0.46, SD=0.15*). The differences in the section timings (as described earlier), may have contributed to differences in output. The relatively large standard deviations for these categories suggest that performance still varies depending on specific interactions. We investigated the videos to better understand other potential reasons for these discrepancies, which we describe as follows:

(i) Differences in available contextual information: The computational model used only a single camera angle for its evaluation. In contrast, the NICE annotations were completed using a composite video comprising three synchronized views: (a) the same single-camera angle

video as used by the model, (b) a Zoom video recording (frontal view), and (c) a screen recording of what students were doing on the computer. These additional perspectives provided contextual cues, such as whether students were leaning in to read from the screen, that sometimes helped the human annotators determine whether a posture constituted a lean or not. This may have led to differences in the interpretation, and consequently, the outputs between the human annotations and the model.

(ii) Differences in criteria for posture change: The model detects leaning based on small changes in spine angle, whereas NICE annotations interpret posture as an indicator of engagement and collaboration, considering posture in relation to other nonverbal behaviors and task context (LaFrance and Broadbent, 1976; Radu, Tu, and Schneider, 2020). As a result, minor postural shifts (e.g., slouching in their seats) may be classified as lean events by the model but not by human annotators.

(iii) Differences in temporality: The model classifies posture every frame (1/60 second), while NICE annotations are not tied to the and require posture changes to be sustained (for approximately 2 seconds) and contextually meaningful. Additionally, brief movements within a sustained lean - either lean in or lean away - (e.g., slight rocking in their chair or adjustment in their seats) were considered as a continuous lean in NICE but were captured as discrete lean-in and lean-out events by the model. As a result, the model detected minor fluctuations that human coders did not, further contributing to discrepancies.

5 Downstream Task Discussion

The 2D perception system integrates multimodal cues such as gaze, posture, gesture, and object interactions captured through audio-visual sensing. These signals can be structured into event streams that represent task progress and social dynamics. Crucially, this continuous monitoring allows the system to identify belief states and

Clip ID	Joint Visual Attention	Look at others	Look away	Individual attention	Point at tool	Point/gesture at others	Lean towards	Lean away
Clip 1	0.36	0.95	1.00	0.18	0.86	0.91	0.55	1.00
Clip 2	0.50	0.88	1.00	0.50	1.00	0.88	0.75	0.88
Clip 3	0.71	0.64	0.93	0.57	1.00	0.93	0.79	0.93
Clip 4	0.88	0.75	0.88	0.63	0.75	1.00	0.88	0.88
Clip 5	0.50	0.71	0.83	0.08	0.79	1.00	0.58	1.00
Clip 6	0.38	0.75	1.00	0.63	0.63	0.75	0.38	0.75
Clip 7	0.63	0.38	0.88	0.63	1.00	1.00	0.13	1.00
Clip 8	0.60	0.47	1.00	0.73	1.00	1.00	0.80	1.00
Clip 9	0.10	0.65	1.00	0.55	0.95	1.00	0.65	1.00
Clip 10	0.17	0.75	1.00	0.42	0.75	0.75	0.83	1.00
Clip 11	0.33	0.17	1.00	0.50	1.00	1.00	0.50	0.83
Clip 12	0.36	0.64	1.00	0.07	1.00	1.00	0.57	1.00
Mean	0.46	0.64	0.96	0.46	0.89	0.93	0.62	0.94
SD	0.22	0.22	0.06	0.22	0.13	0.10	0.22	0.09

Table 0.2: Model accuracy by clip and modality.

Clip ID	Joint Visual Attention	Look at others	Look away	Individual attention	Point at tool	Point/gesture at others	Lean towards	Lean away
Clip 1	0.27	0.82	1.00	0.15	0.66	0.48	0.53	1.00
Clip 2	0.33	0.47	1.00	0.33	1.00	0.47	0.43	0.47
Clip 3	0.65	0.39	0.48	0.36	1.00	0.48	0.63	0.48
Clip 4	0.47	0.43	0.47	0.38	0.43	1.00	0.47	0.47
Clip 5	0.47	0.41	0.45	0.08	0.44	1.00	0.44	1.00
Clip 6	0.27	0.75	1.00	0.62	0.38	0.43	0.37	0.43
Clip 7	0.62	0.27	0.47	0.38	1.00	1.00	0.11	1.00
Clip 8	0.58	0.44	1.00	0.42	1.00	1.00	0.72	1.00
Clip 9	0.10	0.50	1.00	0.35	0.49	1.00	0.50	1.00
Clip 10	0.14	0.43	1.00	0.29	0.43	0.43	0.45	1.00
Clip 11	0.25	0.14	1.00	0.33	1.00	1.00	0.49	0.45
Clip 12	0.26	0.39	1.00	0.07	1.00	1.00	0.36	1.00
Mean (F1)	0.37	0.45	0.82	0.32	0.74	0.77	0.46	0.77
SD	0.18	0.18	0.26	0.15	0.28	0.28	0.15	0.28

Table 0.3: Model F1 scores by clip and modality.

updates during the task process, thereby enabling common ground tracking and friction analysis (Zhu, Jung, et al., 2025; Zhu, Lai, et al., 2025)

In the present study, however, we do not perform low-level feature fusion across modalities. This design choice is motivated by two considerations. First, the NICE framework provides a theoretically grounded, a priori understanding of how different nonverbal modalities relate to collaborative processes, reducing the need for purely data-driven joint feature learning. Second, our current objective is event-based nonverbal indicator prediction rather than joint representation learning.

Accordingly, we focus on detecting and structuring discrete nonverbal events within each modality independently. To support this goal, we adopt a modular architecture that preserves modality-specific outputs. This design allows each modality's predictions to remain interpretable while also enabling their conversion into embeddings at a later stage. In future work, these embeddings—covering both nonverbal and verbal information—can be fused into a unified multimodal representation. This staged approach improves interpretability, facilitates modality-level error analysis, and maintains flexibility for incremental system extension.

In the future, this system can also support semi-automatic detection by extracting verbal and non-verbal behaviors from recorded datasets, assisting human annotators and reducing their workload. Manually annotating nonverbal behaviors is time- and effort-intensive, especially in multi-person, tool-mediated interactions as in collaborative learning settings. By supporting human-in-the-loop annotation, our system reduces manual effort to understand student nonverbal participation during collaboration. This can then be used to inform post-hoc or near-real-time collaborative support in classroom settings.

This detection pipeline can also serve as a critical upstream component for training an intelligent agent. By converting raw audiovisual data into structured, semantic representations—such as recognized gestures or attention shifts—the system provides a high-fidelity state space for reinforcement learning or policy optimization. Instead of learning directly from high-dimensional, noisy pixels, the agent can be trained on these refined behavior streams to make intervention decisions, such as identifying when to intervene to resolve friction or how to maintain common ground based on the group's shared visual focus.

6 Conclusion

We have presented a 2D perception pipeline capable of extracting multimodal cues, including gaze, posture, gesture, facial expression, object interactions, and speech, to capture both verbal and nonverbal behav-

iors in collaborative problem-solving settings. By structuring these signals into coherent event streams, the system provides a foundation for situational awareness and supports downstream applications such as semi-automatic annotation, engagement tracking, and common ground analysis, offering insights into the evolving social and cognitive dynamics of group interaction. Our F1 score results indicate that the model is most effective at detecting rare, visually salient nonverbal behaviors, while struggling with frequent, fine-grained, and context-dependent behaviors such as gaze and subtle posture changes. These findings highlight the need for context-aware modeling and improved temporal and object-level representations when evaluating nonverbal behaviors in collaborative learning settings. As a result, while automated analysis can be useful in saving annotators time and efforts, our study highlights the continued need for human post-processing to ensure accuracy as well as accounting for anomalous behaviors that the model may misinterpret or miss. Future work will extend the model's capabilities to broader domains and further integrate them into AI agents that can function as collaborative partners, enabling real-time reasoning and intervention in multi-party tasks.

Acknowledgments

We are grateful to Kelley Lynch, Ben Lambright, and Bradley Goodman for their valuable assistance with this research. This work was supported in part by the National Science Foundation (NSF) under subcontracts to Brandeis University on award DRL-2019805 (Institute for Student-AI Teaming). It was also supported in part by the U.S. Defense Advanced Research Projects Agency (DARPA) through Transaction award HR00112490377, under the Friction for Accountability in Conversational Transactions (FACT) program.

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